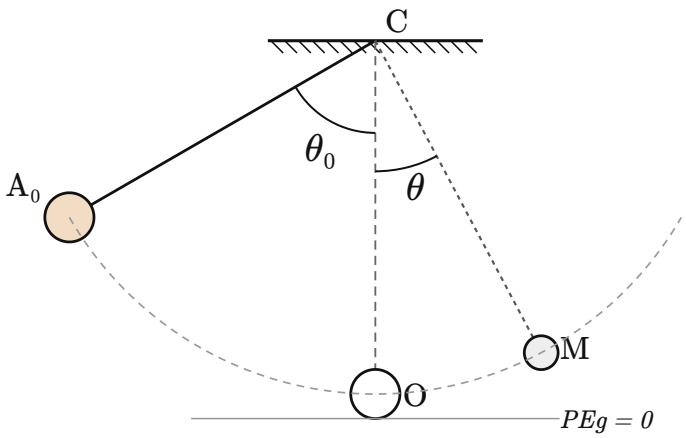
Grade 12 — Life Sciences Lebanese curriculum	**Practice Exam 3** Chapters 1 & 2 — Energy · Linear Momentum	Duration: 2 hours Total: 20 points

This exam is formed of three obligatory exercises. The use of a non-programmable calculator is allowed.

Take, in all exercises: $g = 10\ \text{m/s}^2$.

Exercise 1 — A child on a swing **(7 points)**

A child on a swing is modelled by a simple pendulum: a light inextensible string of length $L = 2.5$ m, fixed at C, carrying a particle of mass $m = 25$ kg. The pendulum is moved away from its equilibrium position O by an angle $\theta_0 = 60°$ (position A_0) and released without initial velocity. At an instant t, the string makes an angle θ with the vertical and the speed of the child is v (Doc. 1).



Doc. 1

> The horizontal plane through O is the reference level of gravitational potential energy.
> $\cos 30° \approx 0.87$; $\cos 53° = 0.6$.

Part 1 — Expressions

1.1 Show that the gravitational potential energy of the system [pendulum, Earth] at M is $GPE = mgL(1 - \cos\theta)$.

1.2 Write the expression of the mechanical energy of the system at M in terms of m, v, g, L and θ.

Part 2 — Ideal swing

In this part, friction is neglected.

2.1 Calculate the mechanical energy of the system at A_0.

2.2 Deduce the speed of the child at O and at the position where $\theta = 30°$.

Part 3 — Real swing

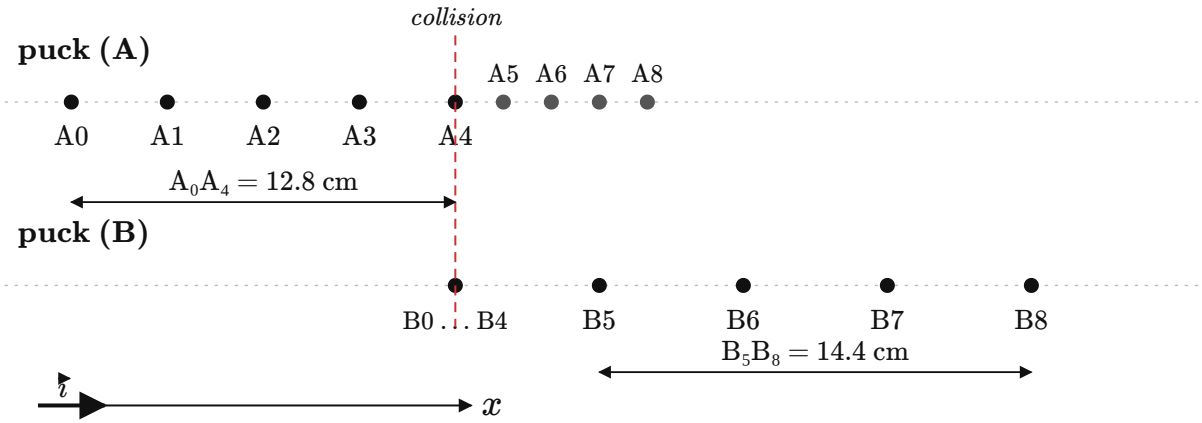
In reality, after passing through O, the child stops on the other side at the angle $\theta_1 = 53°$. This half-oscillation lasts 1.6 s.

3.1 Calculate the variation of the mechanical energy of the system during this half-oscillation.

3.2 Deduce the variation ΔU of the internal energy of the system [swing, air, Earth] and the average power of the friction forces.

Exercise 2 — Dot prints of an elastic collision (7 points)

In a lab session, students study a collision on a horizontal air table. A puck (A) of mass $m_A = 300$ g is launched toward a puck (B) of mass $m_B = 100$ g, initially at rest. The collision is head-on and **perfectly elastic**. A spark recorder marks the positions of the centres of the pucks at successive equal time intervals $\tau = 40$ ms (Doc. 2). Friction is neglected.



Doc. 2 — Positions A_0 … A_4 of (A) before the collision; positions B_5 … B_8 of (B) after it

Part 1 — Reading the recording

1.1 Justify, using Doc. 2, that the motion of (A) before the collision is uniform rectilinear.

1.2 Calculate the speed V_A of (A) before the collision.

Part 2 — Predicting the velocities after the collision

Let $\vec{V}'_A = V'_A\,\vec{i}$ and $\vec{V}'_B = V'_B\,\vec{i}$ be the velocities of (A) and (B) just after the collision.

2.1 Write the two equations that express the conservations during this collision.

2.2 Determine the values of V'_A and V'_B.

Part 3 — Checking with the recording

3.1 Using Doc. 2, calculate the speed of (B) after the collision. Compare it with the value found in 2.2.

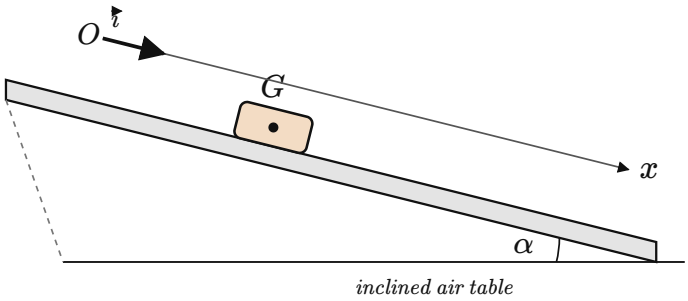
3.2 The contact between the two pucks lasts 20 ms. Calculate the average force exerted by (A) on (B) during the collision.

Exercise 3 — A puck on an inclined air table (6 points)

An air table is inclined by an angle α with the horizontal ($\sin\alpha = 0.1$). A puck of mass $m = 0.5$ kg moves down along the line of greatest slope. During its motion, the puck is acted upon

by a friction force $\vec{f} = -f\,\vec{i}$, constant and opposite to its velocity $\vec{V} = V\,\vec{i}$ ($V > 0$). A recording system gives, at equal time intervals $\tau = 50$ ms, the abscissa x and the speed V of the centre of mass G of the puck along the axis $(O, \vec{i})$ (Doc. 3).

Instant	**Position**	x **(cm)**	V **(m/s)**
$t_0 = 0$	M_0	0	0.20
$t_1 = \tau$	M_1	1.075	
$t_2 = 2\tau$	M_2	2.300	0.26
$t_3 = 3\tau$	M_3	3.675	
$t_4 = 4\tau$	M_4	5.200	0.32
$t_5 = 5\tau$	M_5	6.875	0.35
$t_6 = 6\tau$	M_6	8.700	0.38



Doc. 3

Part 1 — Exploiting the recording

1.1 Calculate the speeds V_1 and V_3 of the puck at the instants t_1 and t_3.

1.2 Calculate the quotient $\dfrac{\Delta P}{\Delta t}$ between t_0 and t_2, then between t_2 and t_6. Compare the two values.

Part 2 — Newton's second law

2.1 Name the forces acting on the puck. Deduce, using Newton's second law, the magnitude F of their resultant.

2.2 By projecting Newton's second law on the axis $(O, \vec{i})$, determine the value of f.

Part 3 — Mechanical energy and friction

The horizontal plane through M_6 is the reference level of gravitational potential energy.

3.1 Calculate the work $W(\vec{f})$ done by the friction force between M_0 and M_6, and the height h between the horizontal planes through M_0 and M_6.

3.2 Calculate the mechanical energy of the system [puck, Earth] at the instants t_0 and t_6.

3.3 Deduce the variation ΔME between t_0 and t_6. Compare it with $W(\vec{f})$.

— End of the exam —